

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**First Semester of B. Pharm Examination****University Theory Examination March 2018****MA121 Basic Mathematics****Date: 31/03/2018, Saturday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 80****Instructions:**

1. There are three sections in this question paper.
2. SECTION -I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION-I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION-I is 30 minutes. Answers to SECTION-I must be written during the first 30 minutes of the examination.
3. Answers to SECTION-II and SECTION-III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I**Q - 1** Choose the correct answer for the following questions each .**[20]**

1. If E and F are finite sets, then $n(E \cup F) + n(E \cap F) =$ _____.
 (A) 0 (C) $n(E) - n(F)$
 (B) $n(E) + n(F)$ (D) $n(E) \times n(F)$
2. If E and F are two sets, then $E \cup (E \cap F) =$ _____.
 (A) E (B) F (C) ϕ (D) $E \cup F$
3. Let $S = \{x | x \text{ is a positive multiple of 3 less than } 20\}$, Then $n(S) =$ _____.
 (A) 6 (B) 5 (C) 7 (D) 3
4. If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $E = \{1, 2, 3, 5\}$, $F = \{2, 4, 6, 7\}$ and $G = \{2, 3, 4, 8\}$, Then $(F \cup G)' =$ _____.
 (A) E (B) $\{1, 5, 9, 10\}$ (C) ϕ (D) $\{2, 3, 4, 6, 7, 8\}$
5. If the function $f : X \rightarrow Y$ is defined for non-empty sets X and Y then,
 (A) X is called domain of f (C) Y is called co - domain of f
 (B) X is called co-domain of f (D) Both (A) and (C).
6. Let the function $f : \mathcal{R} \rightarrow \mathcal{R}$ be defined by $f(x) = 4x - 1$ for all $x \in \mathcal{R}$ then f is _____
 (A) one-one function (C) both (A) and (B)
 (B) onto function (D) neither (A) nor (B)
7. Let the function $f : \mathcal{R} \rightarrow \mathcal{R}$ be defined by $f(x) = 4x + 5$ for all $x \in \mathcal{R}$ then $f^{-1}(x) =$ _____.
 (A) $\frac{1}{4x + 5}$ (B) $\frac{x - 5}{4}$ (C) $-f(x)$ (D) 1
8. The value of $\sin 45^\circ + \cos 45^\circ =$ _____.

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{3}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{3}}{2}$
9. If $\cos A = \frac{4}{5}$ then $\tan A =$ _____
 (A) $\frac{3}{5}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$
10. If the zeroes of the quadratic polynomial $x^2 + (a + 1)x + b$ are 2 and -3, then
 (A) $a = -7, b = -1$ (C) $a = 2, b = -6$
 (B) $a = 5, b = -1$ (D) $a = 0, b = -6$
11. Values of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is _____
 (A) 0 only (B) 4 (C) 8 only (D) both 0 and 8
12. If $x > 0, y > 0$ are real numbers then $\log_{10} xy =$ _____
 (A) $\log_{10} x + \log_{10} y$ (C) $\log_{10} x / \log_{10} y$
 (B) $\log_{10} x - \log_{10} y$ (D) $\log_{10} y / \log_{10} x$
13. $5 \log 2 + \log 3 - \log 8 =$
 (A) 5 (B) $\log 12$ (C) $\log 5$ (D) $5 \log 3$
14. $\lim_{x \rightarrow 5} (x^2 - 4x + 3) =$ _____
 (A) 4 (B) 3 (C) 1 (D) 8
15. The slope of the curve $y = (x + 1)^2$ at point $x = -2$ is _____.
 (A) 2 (B) -2 (C) 3 (D) 0
16. If $y = 3x^4$ then derivative $\frac{dy}{dx} =$ _____.
 (A) 12 (B) x^3 (C) $12x^3$ (D) $4x^3$
17. The integration of the function $\int \frac{1}{2\sqrt{x}} dx =$ _____.
 (A) $\sqrt{x}$ (B) $\sqrt{x} + C$ (C) $\frac{\sqrt{x}}{2} + C$ (D) $x^2 + C$
18. If $P = \begin{bmatrix} 1 & 2 \\ -2 & 0 \end{bmatrix}$, and $Q = \begin{bmatrix} -1 & -2 \\ 2 & 0 \end{bmatrix}$ then $P + Q =$ _____.
 (A) $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$
19. If $U = \begin{bmatrix} 1 & 2 \\ -2 & 0 \end{bmatrix}$ then $|U| =$ _____.
 (A) 0 (B) -4 (C) 4 (D) 5
20. If $P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 \\ 2 & 0 \end{bmatrix}$ then $PQ =$ _____.
 (A) P (B) Q (C) $-Q$ (D) $-P$

SECTION II

Q - 2 Attempt any **Four** of the following.

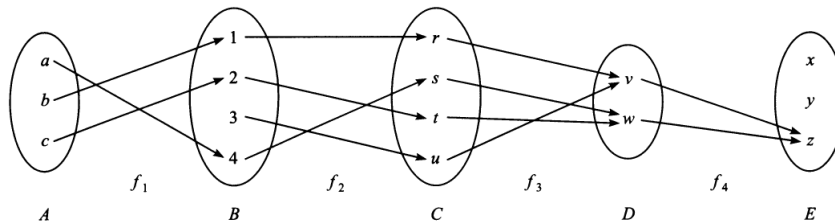
[20]

A. Suppose $n(U) = 70, n(E) = 30, n(F) = 45, n(E \cap F) = 10$. Find

[05]

(a) $n(E \cup F)$ (b) $n(E')$ and $n(F')$ (c) $n(E' \cap F')$

B. Consider the function $f_1 : A \rightarrow B, f_2 : B \rightarrow C, f_3 : C \rightarrow D$ and $f_4 : D \rightarrow E$ shown in Fig below.



Determine whether the functions are (a) One-to-One, (b) Onto.

[05]

C. Let $S = \{a, b, c\}, T = \{b, c, d\}$ then Find $S \times T$.

[05]

D. Evaluate the limits (a) $\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3}$. (b) $\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25}$.

[05]

E. Determine whether the function $f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x \neq 2 \\ 3 & x = 2 \end{cases}$ is continuous at $x = 2$.

[05]

F. Evaluate the limit $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$.

[05]

SECTION III

Q - 3 Attempt any **Four** of the following.

[20]

A. Evaluate the integral $\int (3x^6 - 2x^2 + 7x + 1)dx$.

[05]

B. Evaluate the integral $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sin x dx$.

[05]

C. If $P = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ Find P^{-1} .

[05]

D. Find the derivative of $y = \frac{\sin x}{1 + \cos x}$.

[05]

E. If $P = \begin{bmatrix} 2 & -3 & -4 \\ 1 & 0 & -2 \\ 0 & -5 & -6 \end{bmatrix}$ then compute $|P|$.

[05]

F. Find the first and second derivative of $y = (x^3 - 5)(2x + 3)$.

[05]

Q - 4 Attempt any **Four** of the following.

[20]

A. Solve the following system of equations

$$x + y + z = -1, x + 2y + 3z = -4, x + 3y + 4z = -6 \text{ using Cramer's rule.}$$

[05]

B. (a) Solve the quadratic equation $x^2 + \frac{1}{6}x - 2 = 0$

[02]

(b) Solve the system of equations $x - 7y = 10, 2x - y = 7$ using Cramer's rule.

[03]

C. Verify the identity $\tan x + \cot x = \frac{\operatorname{cosec} x}{\cos x}$.

[05]

D. verify the identity $\frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$.

[05]

E. Use method of substitution and evaluate $\int_0^2 x(x^2 + 1)^3 dx$.

[05]

F. Use method of substitution and evaluate $\int x \sin(2x^2) dx$.

[05]
